Document S1

Supplemental Information

A partitioned, calibrated framework for variant interpretation and differential diagnosis in inherited bleeding and platelet disorders

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Contents

Tables S3, S5, S7 and S9 are supplied as separate Excel files, being large data tables. All supplemental figures and tables are also archived, with the scripts that generate them, in the data record.

Supplemental Methods

Detail supporting the Material and methods section of the main text. Nothing here is required to follow the argument; everything here is required to reproduce it. Every procedure below is implemented in the archived code, and the file that implements it is named at the end of each subsection.

Variant Curation Expert Panel specifications as implemented

Gene-specific rule sets are encoded as versioned YAML, one file per specification, loaded at run time rather than compiled into the scoring logic. Four elements are captured per specification: the frequency thresholds that determine BA1, BS1 and PM2; any modification to a criterion's default strength; the computational thresholds that determine PP3 and BP4; and the identifier of the specification itself, so a classification can be traced to the rule set that produced it.

Frequency criteria. Thresholds were taken from the ClinGen Criteria Specification Registry where a specification exists, and cross-checked against the allele frequencies curators cite in the corresponding Evidence Repository records. This cross-check corrected several values during development and is reported in the main text as 97.8% concordance on 629 variants for which a curator-cited frequency was available.

Strength modification. Where a panel modifies a criterion's default strength, the modification is encoded explicitly. The most consequential is PM2 at Supporting rather than the ACMG default of Moderate, which several panels specify and which materially changes the point total for rare missense variants, PM2 being the most frequently applied criterion in this domain.

Computational thresholds. Each specification carries its own REVEL and SpliceAI cut-offs for PP3 and BP4 where the panel defines them. Where a panel does not, the ClinGen-calibrated bands are used as a fallback. The RUNX1 specification uses the Myeloid Malignancy panel's version 2 values (PP3 at REVEL >= 0.88, BP4 at REVEL <= 0.50) rather than generic defaults.

Residual simplifications, stated. Two elements of the published specifications are not fully encoded. PVS1 is implemented as the Abou Tayoun decision tree rather than as each panel's gene-specific elaboration of it, and PS4 is implemented as a generic odds-ratio and proband-count tree rather than each panel's semi-quantitative counting rules. Both are documented simplifications rather than omissions; neither affects the results reported here, since PS4 is never applied for want of case-control input and PVS1 does not apply to the missense surface that carries the analysis.

Implemented in rules/vcep/loader.py and rules/vcep/specs/.yaml.*

The evidence partition

Each ACMG/AMP criterion is mapped to exactly one owning factor by a static table. Criterion strings are normalised before lookup, so a strength-modified form such as PM2_Supporting resolves to the same owner as PM2. The mapping is total over the criterion vocabulary encountered in the Evidence Repository: the genome-wide analysis reports 100.0000% coverage over 38,802 applied criteria with no unrecognised code.

Exclusivity is enforced by construction rather than by convention, and verified by a test that supplies the joint model a variant with the criteria PP4, PS3, PP1 and PM3 added explicitly and asserts the variant marginal is unchanged to within 1e-12. Because those criteria are owned by other factors, re-adding them to the genetic stream must have no effect; the test fails if the partition is ever weakened.

Implemented in rules/vcep/partition.py; verified by tests/test_discern_joint.py.

The gene term and its two bounds

The joint factorises as

```
P(D, V | E)  proportional to  P(E_pheno | D) . P(E_geno | V) . P(E_func | D, V)
. P(V | G, D) . P(G | D) . P(D)
```

P(V | G, D) normalises over the five variant states, so it cancels when the joint is marginalised to a disease posterior. Without the separate P(G | D) term the gene therefore carries no information about disease identity, which is the defect described in the main text. P(G | D) is implemented as an on-gene/off-gene likelihood pair, giving a likelihood ratio of 4.

Two bounds constrain that value, and both are architectural rather than fitted:

1. It must not exceed a cluster's deciding observation. The sharpest discriminator in the knowledge base is the ristocetin-induced platelet aggregation mixing study, at a likelihood ratio of approximately 11.5. If the gene term exceeded that, no laboratory result could overturn the gene, and the value-of-information layer could never predict a change of leading diagnosis. A ratio of 4 is the largest tested value at which the mixing study still overturns the gene. 2. It must not veto a treatment-divergent competitor. Finding a variant in one gene does not exclude a disease of another gene, which may never have been sequenced or whose variant may be benign. The safety interlock is therefore evaluated on a gene-blind posterior, obtained by omitting P(G | D), while the leading diagnosis reported to the user remains the gene-aware one.

A sweep across the full range of the parameter (Figure S1) shows hard-stop sensitivity and specificity invariant at 100% throughout, the curated diagnosis result flat above the committed value, and the deciding-assay bound failing at ratios of 9 and above.

Implemented in jointdx/factorgraph.py; swept by bench/phase_r_gene_term_sensitivity.py.

Calibration and its out-of-sample estimation

Variant scores are calibrated by isotonic regression. Estimation is out-of-fold throughout: a 5-fold stratified split with shuffling and a fixed seed is constructed once per surface; the isotonic map is fit on the four training folds and applied to the held-out fold; expected calibration error and Brier score are accumulated on held-out predictions only. No variant is ever scored by a calibrator that has seen its label.

Three properties are asserted in code at run time rather than checked afterwards: that no fold trains and evaluates on the same index, that no index appears in two evaluation folds, and that every index is held out exactly once. Any violation raises rather than returning a number.

The fold assignment itself is published (benchmarks/variant_arm/calibration_folds.json in the data record), listing for every variant which fold held it out, together with the full training and evaluation index sets. A reader can verify the three properties directly rather than taking the protocol on trust.

Expected calibration error uses ten equal-width bins over the unit interval, weighting each bin's absolute gap between mean predicted probability and observed frequency by that bin's occupancy. The final bin includes probability 1.0.

Comparator calibration. REVEL and AlphaMissense were put through the identical fold assignment and the identical isotonic procedure, and are reported both raw and calibrated. This is what makes the calibration comparison like-for-like, and it is what caused the calibration-uniqueness claim of an earlier draft to be retired.

Implemented in bench/phase_r_variant.py; the calibration primitive is core/stats.ece.

Statistical procedures

Confidence intervals on proportions. Wilson score intervals. Preferred to the normal approximation because the reported rates sit far from one half at large sample size, where Wald intervals are unreliable.

Confidence intervals on metrics. Percentile bootstrap, 1,000 resamples, fixed seed. Resamples in which a class is absent are discarded rather than scored. Intervals are the 2.5th and 97.5th percentiles of the resampling distribution.

Comparing two areas under the curve. DeLong's test in the fast form of Sun and Xu, computed on the variants both tools score, with ties handled by mid-ranks. Reported alongside a paired bootstrap on the difference, which resamples variants and rescores both tools on the same resample, so the interval reflects the pairing.

Comparing two classifiers on the same cases. Exact McNemar, using the binomial test on the discordant pairs. Chosen over the chi-squared approximation because the discordant counts here are small.

Rank metrics. Recall at k and mean reciprocal rank from one-based hit positions, with a non-hit contributing zero to the reciprocal rank.

Every one of these is implemented once, in a module with no dependency beyond numpy and scipy, and is covered by tests against closed forms and hand-computed values. They were extracted from the analysis harnesses precisely so that the arithmetic behind the reported intervals could be exercised by continuous integration rather than assumed.

Implemented in core/stats.py; verified by tests/test_stats.py.

The ClinVar-blinded protocol, per tool

Several comparators consume ClinVar through the PP5 and BP6 criteria, which would let them reproduce rather than predict the label on any ClinVar-derived truth surface. The protocol therefore differs by tool and is stated for each.

- DISCERN applies no ClinVar-derived criterion. PS1 and PM5, which require a same-residue

ClinVar lookup, are implemented but deliberately not wired in for these runs. This is the reason they appear zero times in the per-criterion comparison, and it is a protocol choice rather than a gap in the engine.

- REVEL and AlphaMissense are precomputed scores and consume no classification.
- GeneBe does apply PP5 and BP6. Rather than exclude it, it is retained as an exhibit: it

attains a perfect area under the curve on the eRepo surface and continues to do so under the time-split, which demonstrates the circularity concretely.

- InterVar and BIAS-2015 are positioned by published metrics obtained on different variant sets,

and are labelled as such wherever they appear.

The time-split panel additionally restricts grading to variants whose expert classification post-dates the ClinVar snapshot bundled with the comparator tools (2021-05-01), so that no tool could have memorised the label irrespective of protocol.

The curated case benchmark and the HPO crosswalk

Cases were curated from published reports, each represented as the causal gene together with the clinical and laboratory findings the report describes, and each carrying the PubMed identifier of its source. No article text is reproduced anywhere in the deposit; the benchmark ships as identifiers, extracted terms, and the expected diagnosis.

Representability in the Human Phenotype Ontology was determined by inverting the committed crosswalk, which maps HPO terms to the engine's internal finding vocabulary, and asking which findings used anywhere in the benchmark have at least one HPO term. Thirteen of the forty-eight distinct findings do. The thirty-five that do not are laboratory assay results with no HPO representation, including the ristocetin mixing study, flow cytometry for CD42 and alphaIIbbeta3, multimer patterns, light transmission aggregometry, and the prothrombinase assay. Nineteen of the forty-two cases carry no HPO term at all and cannot be ranked by a phenotype-driven tool.

Implemented in eval/phenotype_tool_comparison.py; verified by tests/test_diagnosis_baselines.py.

The phenotype-blind baselines

Three reference points, none of which sees a phenotype:

- Gene lookup. Ranks the cluster by placing the diseases that list the case's gene first, each block ordered by prior. A static table with no likelihood ratios and no findings. Verified to produce an identical ranking when every finding is stripped from the case.
- Prior only. Ranks the cluster by prior and ignores the gene entirely.
- Uniform random within cluster. The floor, reported as the analytic expectation together with a simulated interval.

Cases are stratified into three mutually exclusive groups: those whose gene maps to exactly one disease within its cluster, those carrying no causal gene, and the informative subset whose gene is present but maps to more than one disease, so that something other than the gene must break the tie. The three groups partition the benchmark exactly.

Implemented in eval/gene_only_baseline.py.

LIRICAL run configuration

LIRICAL version 2.4.1 was run in phenotype-only mode inside an eclipse-temurin:17-jre container. Nothing was installed on the host.

Per case, the observed HPO terms were supplied via --observed-phenotypes and, where the case records an explicitly absent finding with an HPO representation, the corresponding terms were supplied via --negated-phenotypes, so that LIRICAL receives the same pertinent negatives the engine uses. Genome build was left unset, which selects phenotype-only operation and requires no VCF. Only the twenty-three cases carrying at least one HPO term were run; the remaining nineteen cannot be ranked by a phenotype-driven tool and are reported as such rather than scored as failures.

A practical note for reproduction. LIRICAL's own downloader retrieves mim2gene_medgen over FTP, which many institutional networks block. The same file is served over HTTPS from the identical NCBI path, and substituting it allows the remaining resources to download normally.

Scoring. LIRICAL ranks diseases genome-wide by OMIM and Orphanet identifier, whereas the engine ranks within a cluster of three to eight. Scoring is therefore performed at gene and disease-family resolution: a hit counts if the top-k contains any disease identifier the Human Phenotype Ontology annotates to the case's causal gene. The gene-to-disease crosswalk is derived from the committed HPO annotation file rather than asserted, so every identifier traces to a public source. This is the generous reading for LIRICAL, since it cannot be penalised for differences in OMIM granularity.

A second, like-for-like arm collapses LIRICAL's genome-wide ranking onto the same cluster members the engine ranks, so both tools order the same three to eight diseases. That arm is the one the main text quotes.

Exomiser was not run. It requires a VCF, and seeding one per case with zygosity matched to each report's stated inheritance introduces a correctness risk judged larger than the comparison's marginal value once LIRICAL had established the phenotype-channel result on the same inputs. This is recorded as a scope decision rather than presented as a handicapped run.

Implemented in eval/lirical_arm.py.

Reproducibility

All reported values are produced by scripts archived with the code record and are locked by regression tests that read the committed outputs, so a change to any benchmark that moves a reported number fails the build rather than propagating silently. A further test greps the documentation for claims retired during pre-submission analysis and fails if any is asserted again without its retraction.

Third-party databases are referenced by versioned manifest giving source URL, version or snapshot date, and checksums of the exact files used, rather than redistributed. Software and database versions are listed in Table S9.

Supplemental figure and table legends

Figure S1. Sensitivity of the diagnosis arm to the strength of the gene term. Curated Top-1 accuracy, hard-stop sensitivity and hard-stop specificity across the full range of the P(G|D) likelihood ratio, from an inert gene term (ratio 1) to a ratio of 19. Safety behavior is invariant throughout. The diagnosis result is flat above the committed value of 4, which is the largest ratio at which a cluster's deciding observation can still overturn the gene; shaded regions mark ratios at which it cannot. Related to Figure 4 and the Material and methods.

Figure S2. Treatment-divergence safety interlock, by scenario. Each of five treatment-divergent scenarios against the two behaviors the interlock must exhibit: firing when the planned management is contraindicated for a disease that has not been excluded, and remaining silent when the planned management is harmless. Sensitivity and specificity are both 100%. Adjudication is performed on a gene-blind posterior against the engine's own leading call, so that a variant in one gene cannot silently retire a treatment-divergent disease of another. Related to the trustworthiness results.

Figure S3. Diagnosis-posterior calibration on the curated benchmark. Mean predicted confidence against observed accuracy on the curated case set (n = 42), with the expected calibration error. The shaded band marks the region above the 0.8 confidence threshold, in which no incorrect call was made. Reported as a proof-of-concept characterization: with the diagnosis arm at ceiling on this benchmark the calibration figure reflects mild underconfidence rather than a discriminating result, and the abstention layer's operational value requires a benchmark that is not at ceiling. Related to the withdrawn risk-coverage claim.

Figure S4. Independent sensitivity on the CDC hemophilia mutation projects. Likely-pathogenic or pathogenic recall on null variants in the CHAMP and CHBMP catalogs (2,130 null variants of 5,437 disease alleles), scored by consequence alone with no frequency or predictor input. The lower F9 figure reflects that gene's large terminal exon, where premature termination codons escape nonsense-mediated decay and are correctly held at uncertain significance rather than called pathogenic. Related to the independent sensitivity result.

Figure S5. A worked example: the safety interlock on a single case. (A) The evidence supplied: a GP1BA missense variant absent from gnomAD, enhanced low-dose ristocetin-induced platelet aggregation, a mixing study of platelet origin, plasma origin explicitly excluded, thrombocytopenia, and desmopressin as the planned therapy. (B) The criterion trail. PM2 is applied and owned by the variant factor. PP4 and PS3 are owned by the disease and functional factors and are not available from sequence; PM5 and PS4 are variant-intrinsic but have no input under this protocol. The variant therefore remains of uncertain significance. (C) The ranked differential, in which platelet-type von Willebrand disease leads at 99.3%. (D) The decision output. Desmopressin is contraindicated in type 2B von Willebrand disease, a VWF disease, whereas the variant here is in GP1BA. Because safety is adjudicated gene-blind, type 2B retains 0.9% in the safety view and the hard stop fires, together with the sequencing test that would resolve the question. This is the framework's clinical argument on one case: a low-probability but treatment-divergent competitor is not silently retired by the gene. Related to Figure 1.

Table S1. Per-criterion agreement with expert-panel applications. Every ACMG/AMP criterion encountered on the Evidence Repository surface, with the number of times the expert panels applied it, the number of times DISCERN applied it, Cohen's kappa where both applied it at least once, and, for criteria applied zero times, the reason: routed to another factor by the evidence partition, or variant-intrinsic with no input under this protocol.

Table S2. Attribution of the intrinsic-evidence ceiling. Variants reaching Likely Pathogenic under each restoration condition: as scored; restoring the variant-intrinsic criteria this pipeline cannot annotate; restoring the criteria the partition routes to other factors; and restoring both. Followed by the reason each unapplied criterion has no input, classified as a protocol choice, a data-availability limit, or an engine scope gap. Counts use default criterion strengths and are therefore lower bounds.

Table S3. Discrimination likelihood ratios for the ten confusable-disease clusters. Every likelihood ratio in the disease-discrimination model, given as the probability of the finding under each disease, with the sample size and the PubMed identifier of the primary source. Continuous integration fails if any entry lacks a source. Supplied as a separate Excel file.

Table S4. Treatment-divergence scenarios. Each safety scenario with the disease pair, the planned management, the expected behavior, and the observed behavior, for both the contraindicated and the harmless plan.

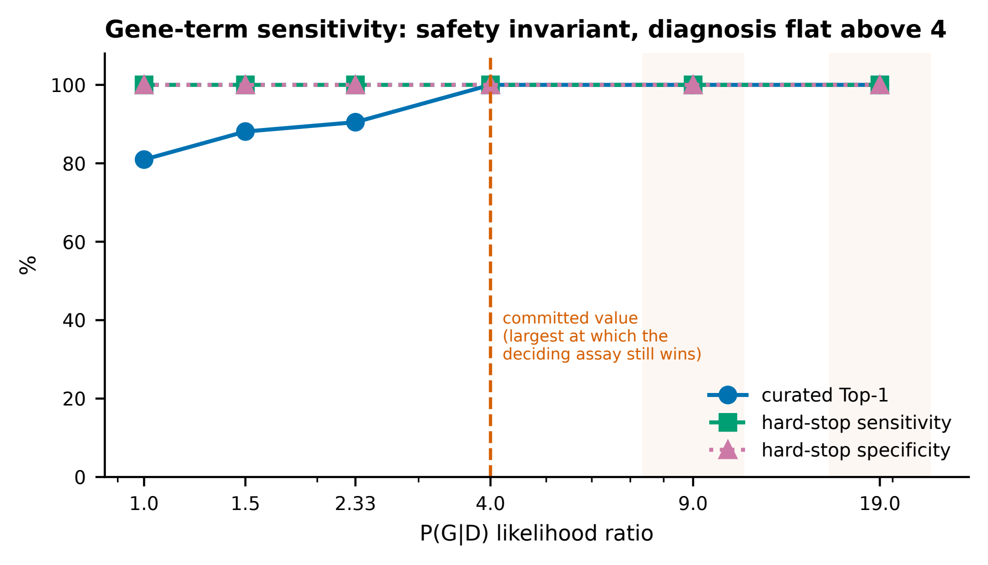
Table S5. The curated published-case benchmark. All 42 cases with PubMed identifier, causal gene, cluster, expected diagnosis, extracted Human Phenotype Ontology terms, stratum, and the leading call and Top-1 outcome for DISCERN and for the phenotype-blind gene lookup. No article text is reproduced. Supplied as a separate Excel file.

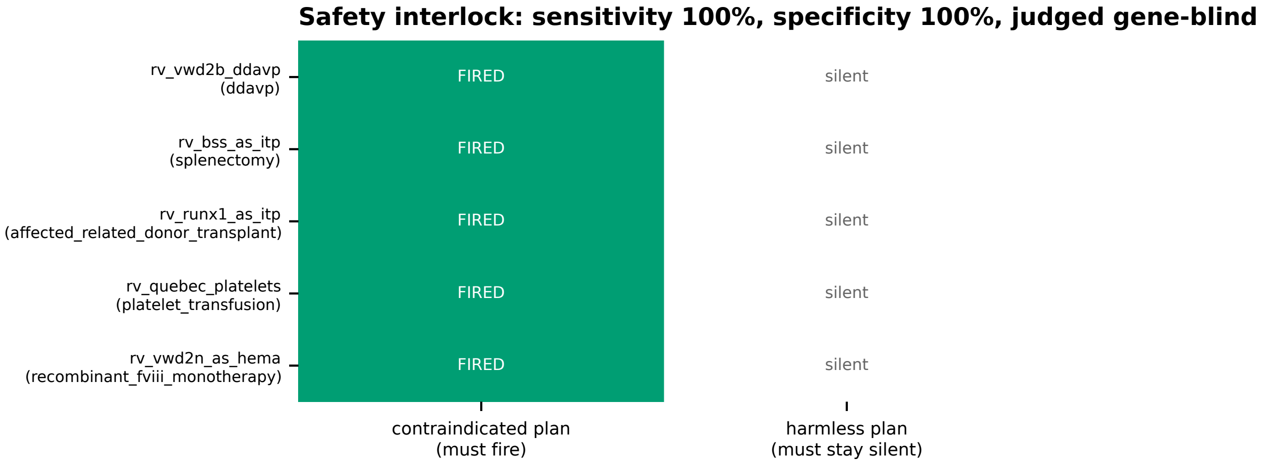
Table S6. CHAMP and CHBMP recall by gene. Likely-pathogenic or pathogenic recall on null variants for F8, F9, and the combined catalogs, with the explanation of the F9 result.

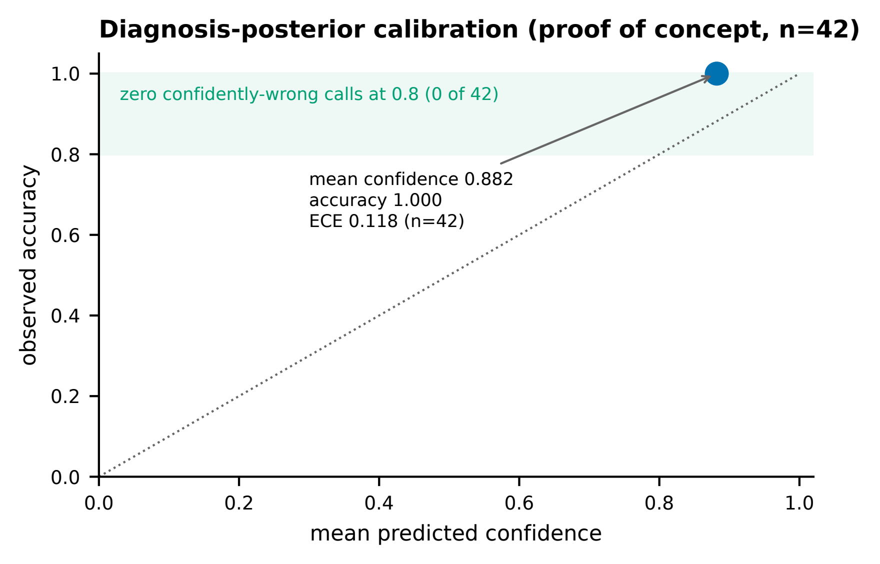
Table S7. Third-party data sources. Each source with the files used, the access URL, the version or snapshot date, and its role. None is redistributed; checksums for the exact files used appear in the deposit manifest. Supplied as a separate Excel file.

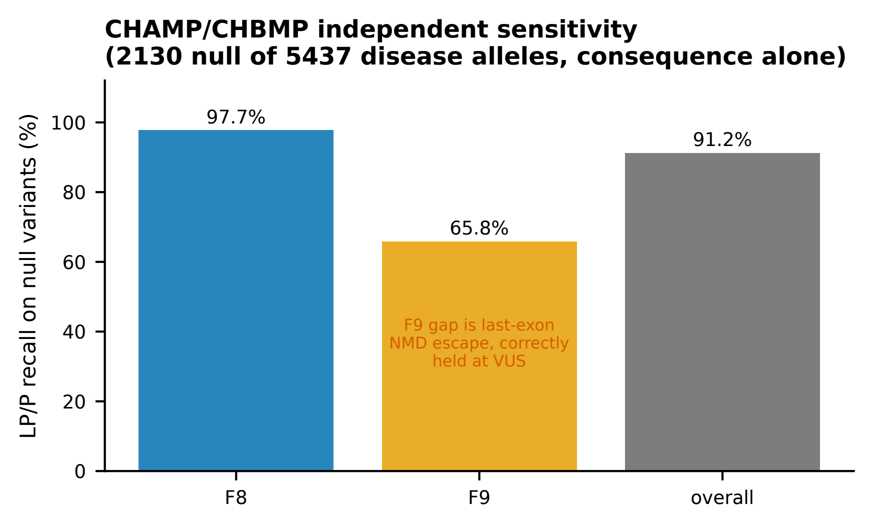
Table S8. LIRICAL comparison arms. Every arm of the external comparison with its sample size, recall at 1, 3 and 5, and mean reciprocal rank, followed by the paired tests against LIRICAL restricted to the same cluster. The hpo_representable_only arm supports claims about inference on identical evidence; the phenotype_only arm supports the architectural claim about what the model can represent. The post-correction arm is in-sample and is not a head-to-head result.

Table S9. Software, tool, and database versions. Every component used to produce the reported results, with its version and role. DISCERN's own commit hash is the authoritative version identifier. Supplied as a separate Excel file.


Gene-term sensitivity: safety invariant, diagnosis flat above 4
%
100
80
60
40
20
0
1.0
1.5
2.33
4.0
9.0
19.0
P(G|D) likelihood ratio
committed value
(largest at which the
deciding assay still wins)
curated Top-1
hard-stop sensitivity
hard-stop specificity


Safety interlock: sensitivity 100%, specificity 100%, judged gene-blind
rv_vwd2b_ddavp (ddavp)
rv_bss_as_itp (splenectomy)
rv_runx1_as_itp (affected_related_donor_transplant)
rv_quebec_platelets (platelet_transfusion)
rv_vwd2n_as_hema (recombinant_fviii_monotherapy)
FIRED
FIRED
FIRED
FIRED
FIRED
silent
silent
silent
silent
silent
contraindicated plan (must fire)
harmless plan (must stay silent)


Diagnosis-posterior calibration (proof of concept, n=42)
zero confidently-wrong calls at 0.8 (0 of 42)
mean confidence 0.882
accuracy 1.000
ECE 0.118 (n=42)
observed accuracy
mean predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0


CHAMP/CHBMP independent sensitivity
(2130 null of 5437 disease alleles, consequence alone)
LP/P recall on null variants (%)
0
20
40
60
80
100
97.7%
65.8%
91.2%
F9 gap is last-exon NMD escape, correctly held at VUS
F8
F9
overall

A Evidence supplied

gene	GP1BA
variant	missense, absent from gnomAD
low-dose RIPA	enhanced
RIPA mixing study	platelet origin
plasma origin	**ABSENT (pertinent negative)**
platelet count	thrombocytopenia
planned therapy	**desmopressin (DDAVP)**

B Criterion trail and factor ownership

applied

PM2 -> variant intrinsic

not applied here

PP4 -> disease pp4 (phenotype specificity)

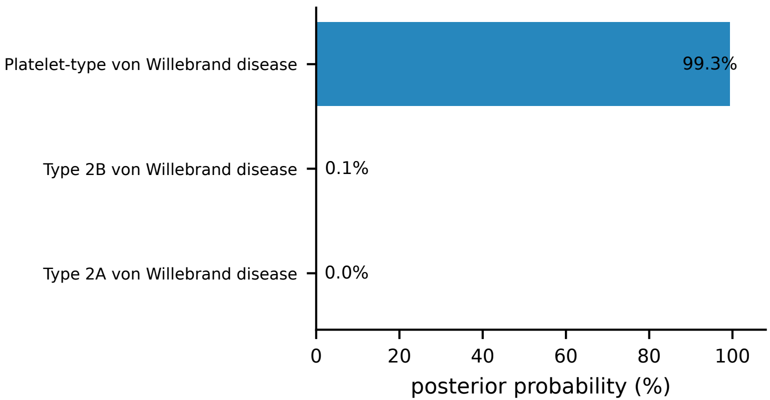
PS3 -> functional (functional assay)

PM5 -> variant intrinsic (same residue, ClinVar-blinded)

PS4 -> variant intrinsic (case-control counts)

variant remains VUS on intrinsic evidence

C Ranked differential



D Decision output

HARD STOP

desmopressin is contraindicated if type 2B von Willebrand disease, which the gene-blind safety view still gives 0.9% and has not excluded

leading call

Platelet-type von Willebrand disease (99.7%), decided

variant classification

uncertain significance - abstained

recommended next observation

Targeted GP1BA vs VWF sequencing

Supplemental tables

Table S1. Per-criterion agreement with expert-panel applications

ACMG_criter	erepo_appl	discern_appl	kappa	both_appl	strength_agree	zero_application_cause	note
PVS1	178	173	0.977	172	0.041	applied by both	applied at Moderate here ag
PP3	305	293	0.947	287	0.969	applied by both	derived from the same predi
BA1	183	98	0.64	95	1.0	applied by both	frequency-based; DISCERN is
BS1	85	58	0.297	24	1.0	applied by both	frequency-based; DISCERN is
PM2	812	479	0.084	336	0.94	applied by both	applied more liberally than
BP4	615	91	0.074	67	1.0	applied by both	applied more sparingly than
BP7	541	0		0		not implemented	applies to synonymous varia
BS2	5	0		0		not implemented	requires observation in hea
PM1	53	0		0		variant-intrinsic; no input	hotspot and critical-domain
PM5	97	0		0		variant-intrinsic; no input	requires a same-residue Cli
PS1	7	0		0		variant-intrinsic; no input	requires a same-residue Cli
PS3	131	0		0		routed to another factor by	routed to the functional fa
PS4	173	0		0		variant-intrinsic; no input	requires case-control count

NOTE: Criteria applied by neither the expert panels nor DISCERN on this surface are omitted (PS2, PM3, PM4, PM6, PP1, PP2, PP4, PP5, BP1, BP2, BP3, BP5, BP6), so these thirteen rows are not the whole of ACMG/AMP. Agreement is reported at two levels: kappa asks whether the criterion was applied at all, and strength_agreement asks, among variants where both applied it, how often they applied it at the same ClinGen strength. An unmodified code is resolved to its criterion's default strength before comparison.

Table S2. Attribution of the intrinsic-evidence ceiling

restoration_condition	variants_reaching_LP	of_n
as scored (intrinsic evidence only)	0	425
+ intrinsic criteria with no input here	27	425
+ criteria the partition routes away	52	425
both restored	89	425

code	reason unapplied	kind of limit
PS1	implemented (adapters/clinvar.py) but d	protocol choice (ClinVar-blinded)
PM5	same as PS1 - implemented, and withheld	protocol choice (ClinVar-blinded)
PS4	implemented as a decision tree in rules	data availability
PM1	genuinely not implemented: no scorer em	engine scope gap

NOTE: Points for the restored codes use default ACMG strengths, because stripping a strength suffix is how the code vocabulary is compared. VCEPs frequently apply gene-specific strengths above the default, so every count here is a lower bound on what the expert evidence would actually contribute.

Table S4. Treatment-divergence scenarios

scenario	planned_management	expected_on_contraind	observed_on_contraind	expected_on_har	observed_on_har
rv_vwd2b_ddavp	ddavp	fire	fired	stay silent	silent
rv_bss_as_itp	splenectomy	fire	fired	stay silent	silent
rv_runx1_as_itp	affected_related_donor_tra	fire	fired	stay silent	silent
rv_quebec_plate	platelet_transfusion	fire	fired	stay silent	silent
rv_vwd2n_as_hem	recombinant_fviii_monother	fire	fired	stay silent	silent

NOTE: sensitivity 100%, specificity 100% over 5 scenarios; adjudicated gene-blind against the engine's own leading call.

Table S6. CHAMP and CHBMP recall by gene

gene	LP_P_recall_on_null	note
F8	0.977	NMD-triggering nulls resolved by PVS1
F9	0.658	gap is the large terminal exon: last-exon PTCs escape NMD and are
overall	0.912	2130 null variants of 5437 disease alleles

NOTE: Recall by consequence alone, with no frequency or predictor input. Source: docs/DISCERN_CHAMP_CHBMP_Benchmark_v1.md

Table S8. LIRICAL comparison arms

arm	n	recall@1	recall@3	recall@5	MRR
LIRICAL_genome_wide	23	0.1304	0.2609	0.3478	0.215
LIRICAL_restricted_to_cluster	23	0.5652	0.9565	0.9565	0.7536
DISCERN_hpo_representable_only	23	0.7391	1.0	1.0	0.8623
DISCERN_phenotype_only_no_gene	23	0.913	1.0	1.0	0.9565
DISCERN_pre_gene_term_fix	23	0.8261	1.0	1.0	0.913
DISCERN_post_fix_IN_SAMPLE	23	1.0	1.0	1.0	1.0

paired test	arm	delta	CI95_low	CI95_high	McNemar_p
vs LIRICAL restricted	phenotype_only	0.3478	0.1304	0.5652	0.021484
vs LIRICAL restricted	hpo_representable_only	0.1739	-0.0446	0.4348	0.289062

phenotype-only arm is invariant to the gene True

NOTE: LIRICAL 2.4.1, phenotype-only mode, observed and negated HPO terms supplied, run in a container. Quote the hpo_representable_only arm for inference claims and the phenotype_only arm for the architectural claim.